

An Asymptotic-Dimension-Zero Counterexample for Asymptotic Coarse Lipschitz Equivalence

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Source Question

The source is Bruno M. Braga and Gilles Lancien, *Asymptotic coarse Lipschitz equivalence*, arXiv:2302.12016. The paper introduces asymptotic coarse Lipschitz equivalence: metric spaces X and Y are asymptotically coarse Lipschitz equivalent if there are coarse Lipschitz maps $f : X \rightarrow Y$ and $g : Y \rightarrow X$ such that gf and fg are asymptotically close to the corresponding identity maps.

After constructing examples separating asymptotic coarse Lipschitz equivalence from coarse equivalence in positive asymptotic dimension, the paper asks:

Problem 2.15. Let X and Y be asymptotically coarse Lipschitz equivalent metric spaces with asymptotic dimension zero. Does it follow that X and Y are coarse (Lipschitz) equivalent?

It is easy to show that this is a coarse property, i.e., it is preserved by coarse equivalences. But while X in the previous example has it, Y does not.

Notice that both restrictions in the previous proposition and in the previous remark only work for metric spaces with asymptotic dimension at least 1 (indeed, the existence of the paths in the remark above implies either that X is bounded or that X has asymptotic dimension at least 1). Moreover, Example 2.9 falls in the same scenario. This justifies the following problem.

Problem 2.15. Let X and Y be asymptotically coarse Lipschitz equivalent metric spaces with asymptotic dimension zero. Does it follow that X and Y are coarse (Lipschitz) equivalent?

Answer

The answer is negative as stated. We construct two metric spaces X and Y with asymptotic dimension zero which are asymptotically coarse Lipschitz equivalent but are not even coarsely equivalent.

Definitions Used

For maps $u, v : Z \rightarrow W$, write $u \sim_\infty v$ if, for some equivalently every base point $z_0 \in Z$,

$$\lim_{z \rightarrow \infty} \frac{d_W(u(z), v(z))}{d_Z(z, z_0)} = 0.$$

A map $h : Z \rightarrow W$ is coarse Lipschitz if its modulus of continuity is bounded above by an affine function. A pair of coarse Lipschitz maps $f : X \rightarrow Y$, $g : Y \rightarrow X$ witnesses asymptotic coarse Lipschitz equivalence if $gf \sim_\infty \text{Id}_X$ and $fg \sim_\infty \text{Id}_Y$.

We use the standard characterization of asymptotic dimension zero: a metric space has asymptotic dimension zero if, for every $r > 0$, the r -chain components have uniformly bounded diameter. In the proof below this condition is verified directly.

Proof Intuition

Start with a very sparse countable metric space $X = \{x_n\}$, where different points are separated by distances on the order of $2^n + 2^m$. Build Y by replacing x_n with an uncountable discrete fiber of diameter n . The fiber diameter n is negligible compared with the distance 2^n from the base point, so collapsing each fiber gives an asymptotic coarse Lipschitz inverse. But a genuine coarse equivalence would have to coarsely cover each far fiber within one fixed radius. For a fiber whose internal distance is larger than that radius, every small ball is a singleton, so a countable space cannot coarsely cover an uncountable fiber.

The Counterexample

Theorem 1. *There exist metric spaces X and Y such that*

$$\text{asdim } X = \text{asdim } Y = 0,$$

X and Y are asymptotically coarse Lipschitz equivalent, but X and Y are not coarsely equivalent. In particular, they are not coarse Lipschitz equivalent.

Proof. Let A be an uncountable set and fix $a_0 \in A$. Let

$$X = \{x_n : n \in \mathbb{N}\}$$

with metric

$$d_X(x_n, x_m) = \begin{cases} 0, & n = m, \\ 2^n + 2^m, & n \neq m. \end{cases}$$

Let

$$Y = \mathbb{N} \times A$$

with metric

$$d_Y((n, a), (m, b)) = \begin{cases} 0, & (n, a) = (m, b), \\ n, & n = m \text{ and } a \neq b, \\ 2^n + 2^m, & n \neq m. \end{cases}$$

The triangle inequality is immediate: cross-fiber distances are star distances $2^n + 2^m$, while an internal fiber distance n is at most $2^n + 2^n$, so passing through another fiber cannot shorten it.

We first check asymptotic dimension zero. Fix $r > 0$, and choose N so large that

$$N > r \quad \text{and} \quad 2^N > r.$$

In X , all points x_n with $n \geq N$ are isolated in the r -chain relation, while the remaining points form only finitely many r -components, all with bounded diameter. Thus X has asymptotic dimension zero.

The same argument applies to Y . If $n \geq N$, then distinct points in the fiber $\{n\} \times A$ are at distance $n > r$, and points in different sufficiently large fibers are also more than r apart. All possible nontrivial r -components lie over the finite set $\{1, \dots, N-1\}$, hence have diameter bounded by a finite constant depending only on r . Therefore Y has asymptotic dimension zero.

Define

$$f : X \rightarrow Y, \quad f(x_n) = (n, a_0),$$

and

$$g : Y \rightarrow X, \quad g(n, a) = x_n.$$

The map f is an isometric embedding, and g is 1-Lipschitz: distances between different fibers are preserved, while distances inside a fiber are collapsed to zero. Hence both maps are coarse Lipschitz. Moreover,

$$gf = \text{Id}_X.$$

For $y = (n, a) \in Y$,

$$d_Y(fg(y), y) = d_Y((n, a_0), (n, a)) \leq n.$$

Taking $y_0 = (1, a_0)$, we have $d_Y((n, a), y_0) = 2^n + 2$ whenever $n \neq 1$. Thus

$$\frac{d_Y(fg(n, a), (n, a))}{d_Y((n, a), y_0)} \leq \frac{n}{2^n + 2} \rightarrow 0$$

as $y \rightarrow \infty$. Hence $fg \sim_\infty \text{Id}_Y$, and f, g witness an asymptotic coarse Lipschitz equivalence between X and Y .

It remains to prove that X and Y are not coarsely equivalent. Suppose, toward a contradiction, that there are coarse maps $F : X \rightarrow Y$ and $G : Y \rightarrow X$ such that

$$FG \sim \text{Id}_Y.$$

Then there is $C < \infty$ such that

$$d_Y(FG(y), y) \leq C \quad (y \in Y).$$

Choose $n > C$. For every $a \in A$, the closed C -ball around (n, a) is the singleton $\{(n, a)\}$: any other point in the same fiber is at distance $n > C$, and every point in a different fiber is even farther away. Therefore

$$FG(n, a) = (n, a) \quad (a \in A).$$

This implies that the image $F(X)$ contains the whole fiber $\{n\} \times A$. But X is countable, so $F(X)$ is countable, whereas $\{n\} \times A$ is uncountable. This is impossible. Hence no coarse equivalence exists. $\square$

Novelty and Duplicate Checks

Before promotion, I searched the cheap run indexes for arXiv 2302.12016, the title *Asymptotic coarse Lipschitz equivalence*, and core phrases including *asymptotic coarse Lipschitz*, *asymptotic dimension zero*, *coarse Lipschitz equivalent*, and Problem 2.15. No existing packet or attempt answered this problem.

I also ran bounded web searches for exact and near-exact phrases including

- "Asymptotic coarse Lipschitz equivalence" "Problem 2.15",
- "asymptotic coarse Lipschitz equivalent" "asymptotic dimension zero",
- "asymptotically coarse Lipschitz equivalent" "coarse Lipschitz equivalent".

No later answer surfaced in this search; the only direct hit was the source paper itself.

Scope and Limitations

The counterexample answers Problem 2.15 as written for arbitrary metric spaces. It is nonseparable and uses uncountable fibers. It does not settle strengthened variants in which the metric spaces are required to be countable, separable, uniformly locally finite, of bounded geometry, or otherwise cardinal-controlled.

References

References

- [1] B. M. Braga and G. Lancien, *Asymptotic coarse Lipschitz equivalence*, arXiv:2302.12016, 2023.

Human Review Recommendation

Review should focus on whether the source problem intended hidden separability or bounded geometry assumptions. Without such additional hypotheses, the construction gives a direct negative answer: the asymptotic error is sublinear along each uncountable fiber, while coarse equivalence would force a countable image to cover an uncountable far fiber at bounded scale.